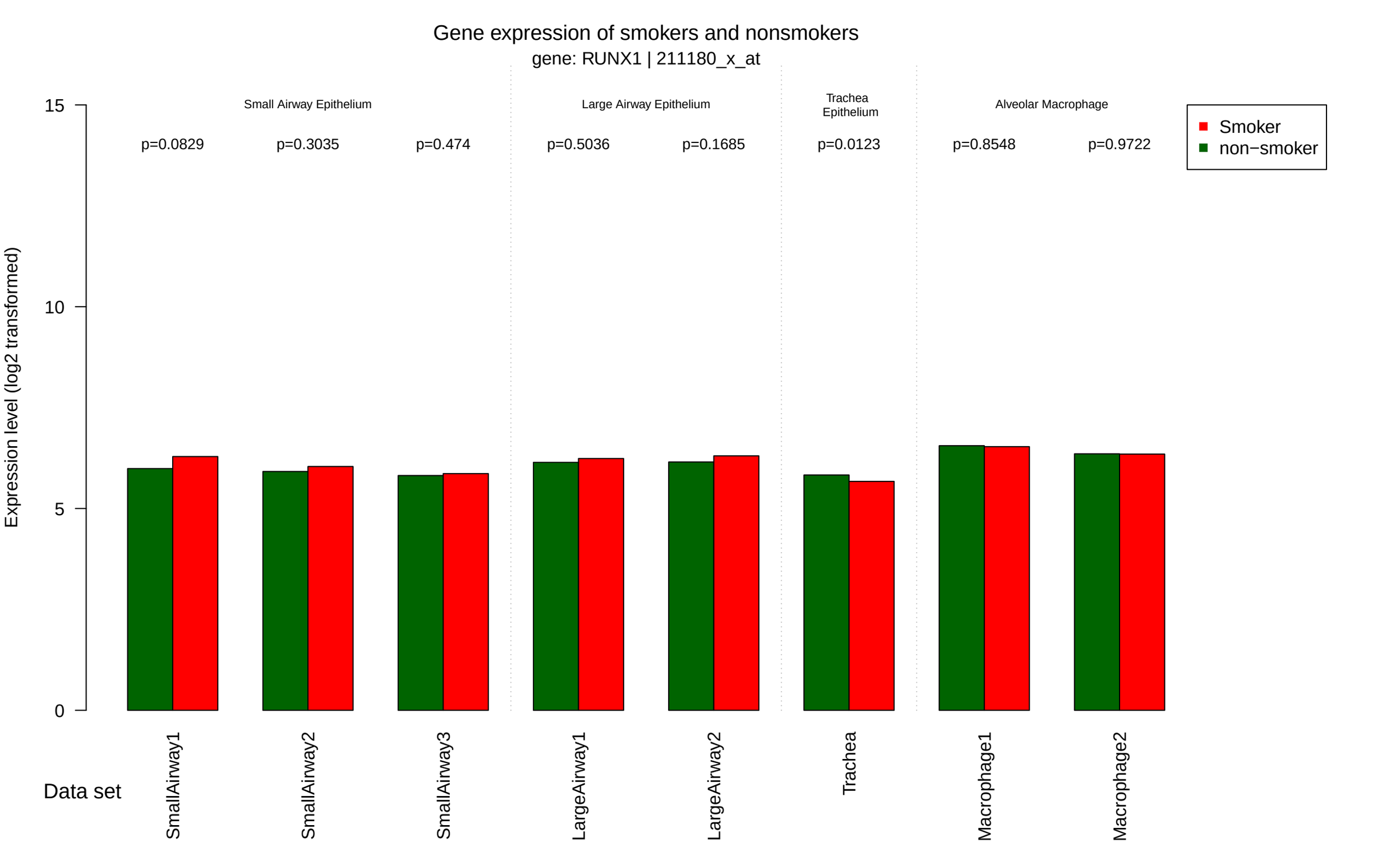
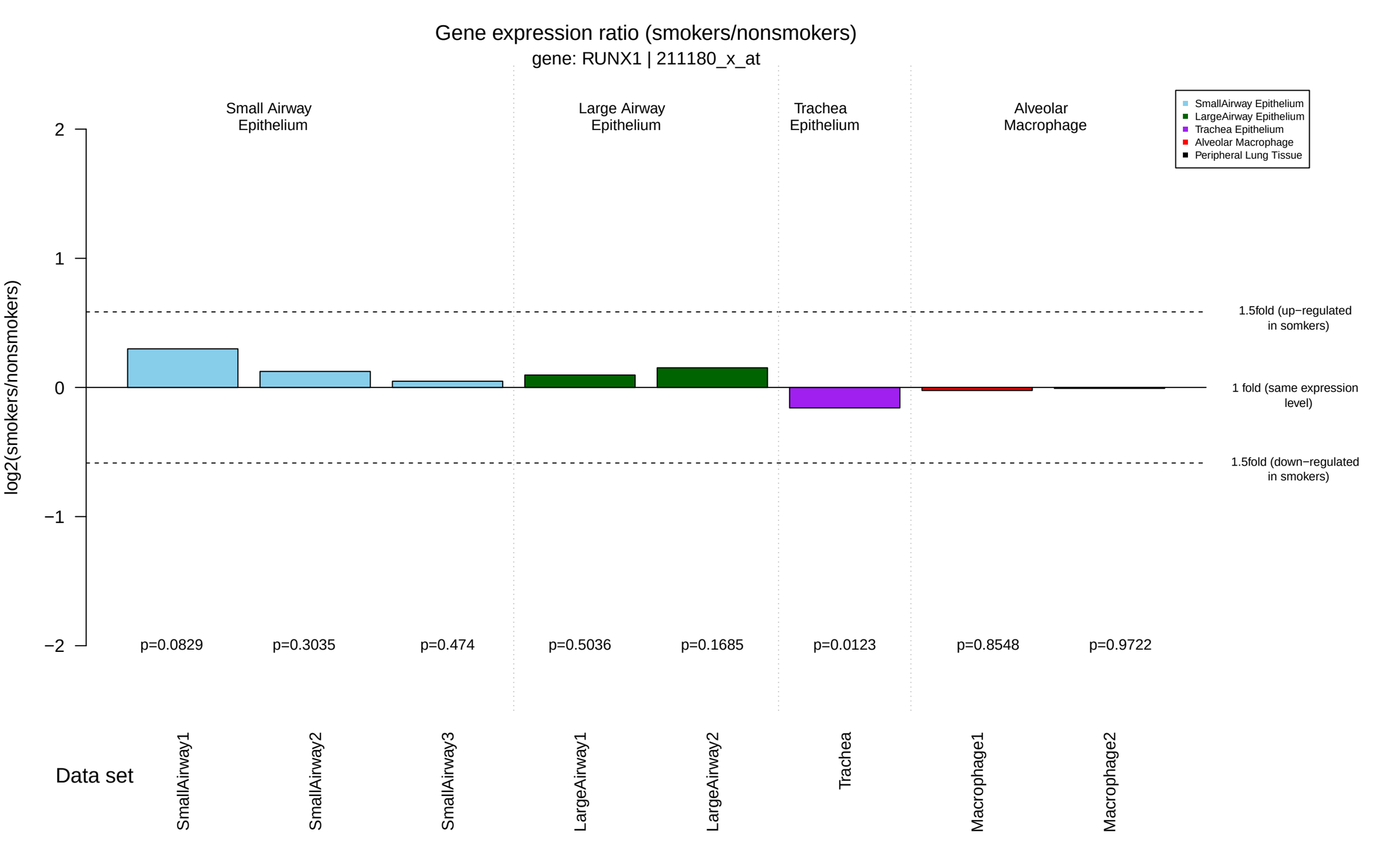
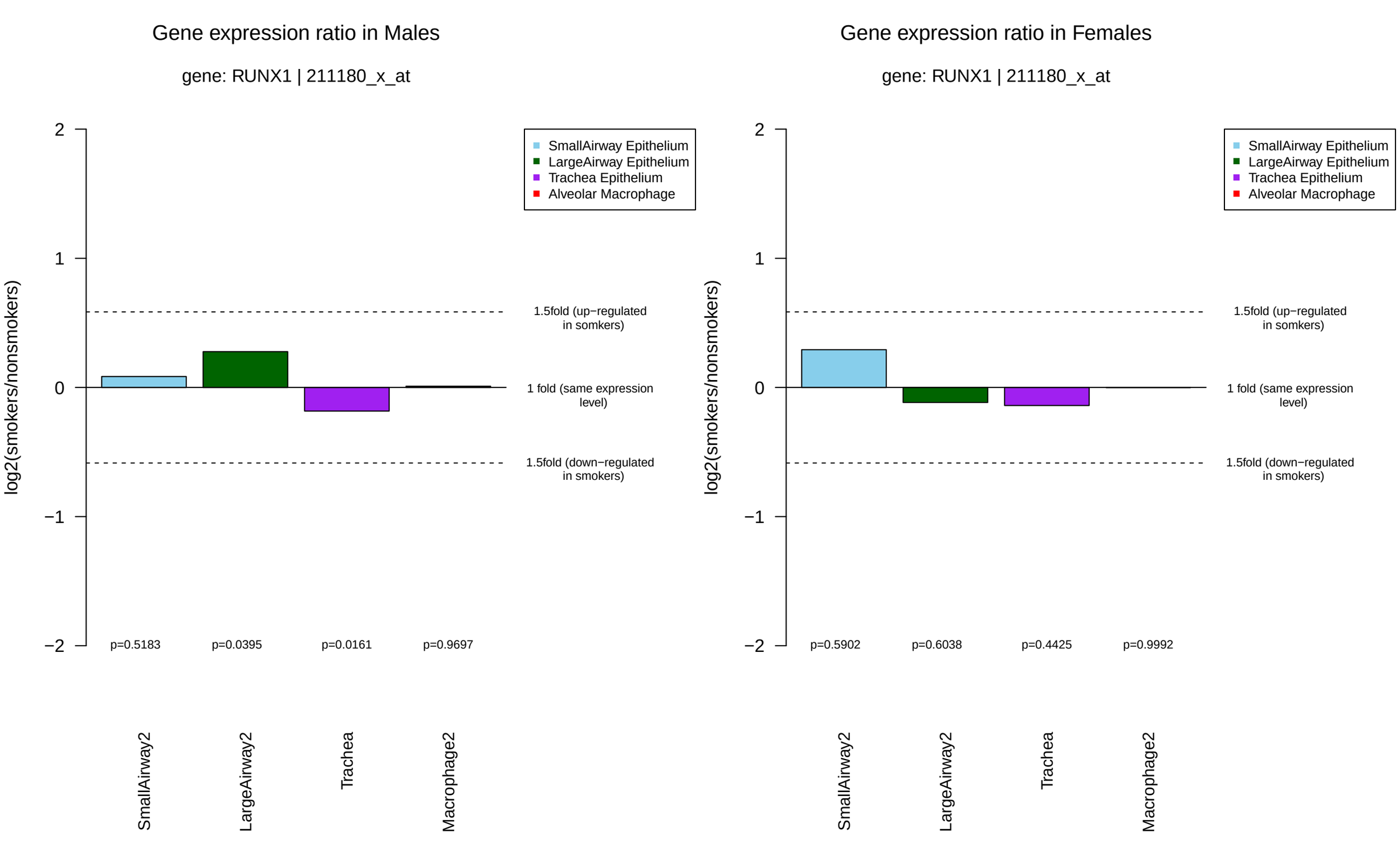
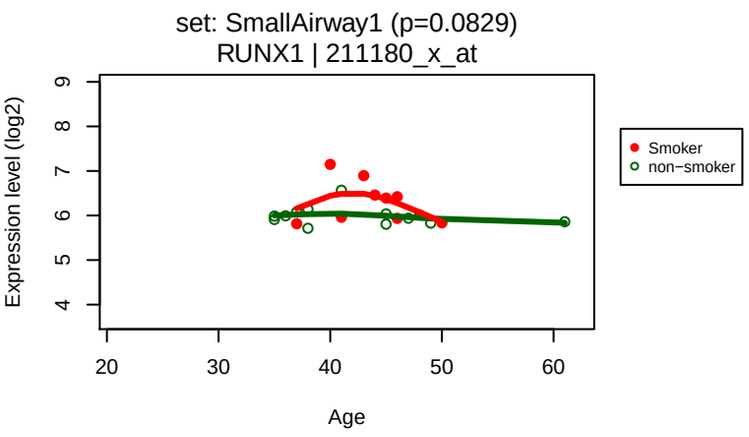
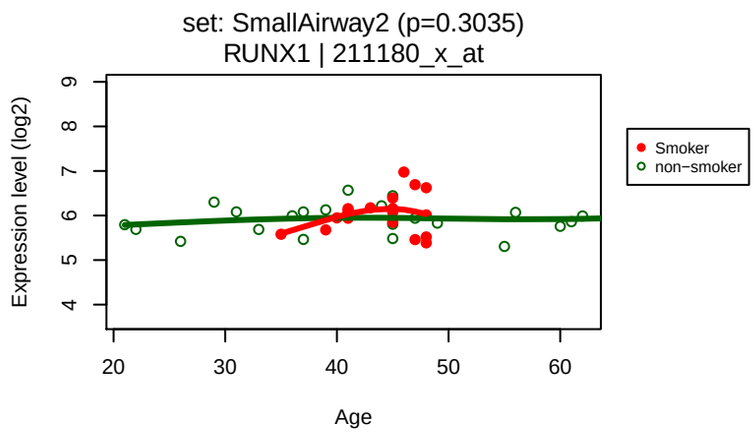
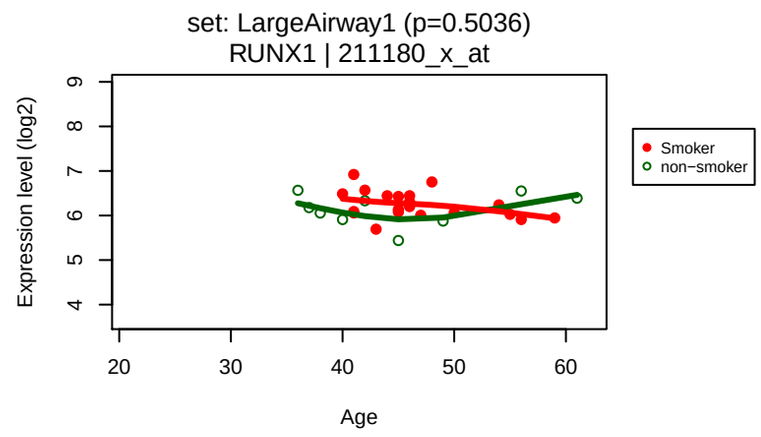

Gene expression of smokers and nonsmokers
gene: RUNX1 | 211180_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0829
p=0.3035
p=0.474
p=0.5036
p=0.1685
p=0.0123
p=0.8548
p=0.9722
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

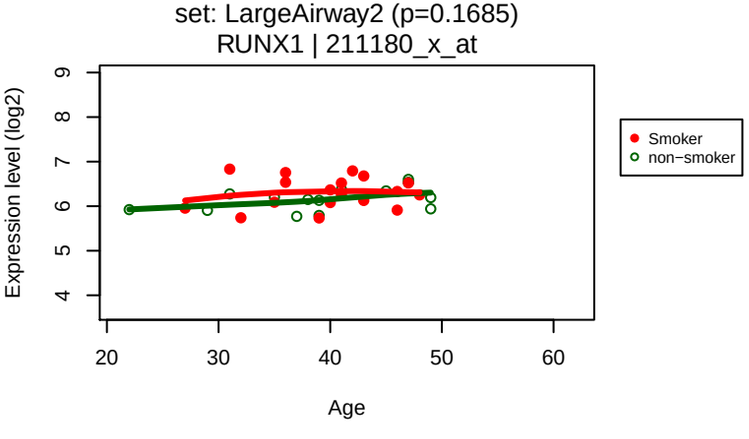

Gene expression ratio (smokers/nonsmokers)
gene: RUNX1 | 211180_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0829
p=0.3035
p=0.474
p=0.5036
p=0.1685
p=0.0123
p=0.8548
p=0.9722
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

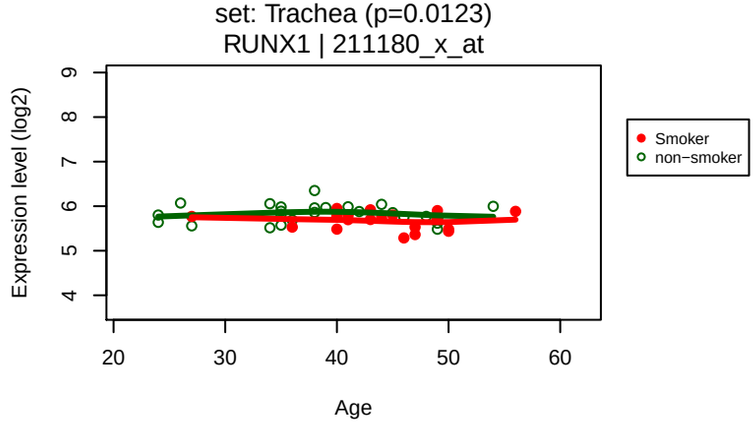

Gene expression ratio in Males
gene: RUNX1 | 211180_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5183
p=0.0395
p=0.0161
p=0.9697
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RUNX1 | 211180_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5902
p=0.6038
p=0.4425
p=0.9992
SmallAirway2
LargeAirway2
Trachea
Macrophage2

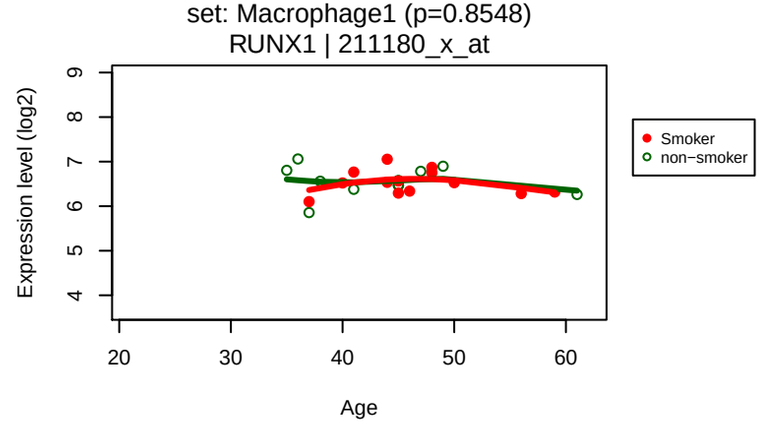

set: SmallAirway1 (p=0.0829)
RUNX1 | 211180_x_at
Expression level (log2)
Age
Smoker
non−smoker

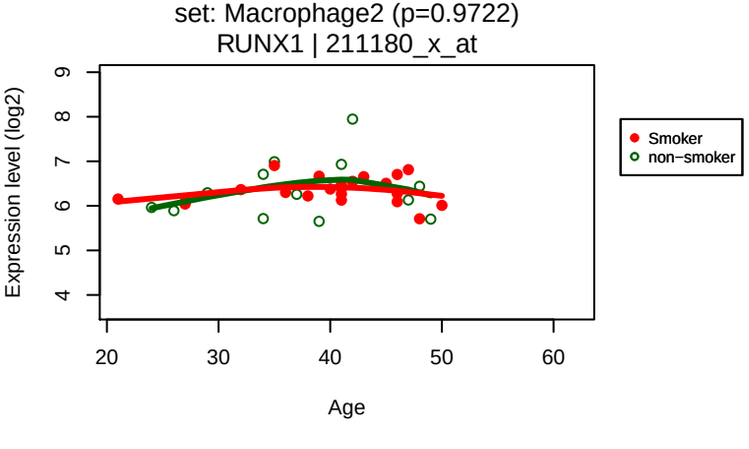

set: SmallAirway2 (p=0.3035)
RUNX1 | 211180_x_at
Expression level (log2)
Age
Smoker
non−smoker

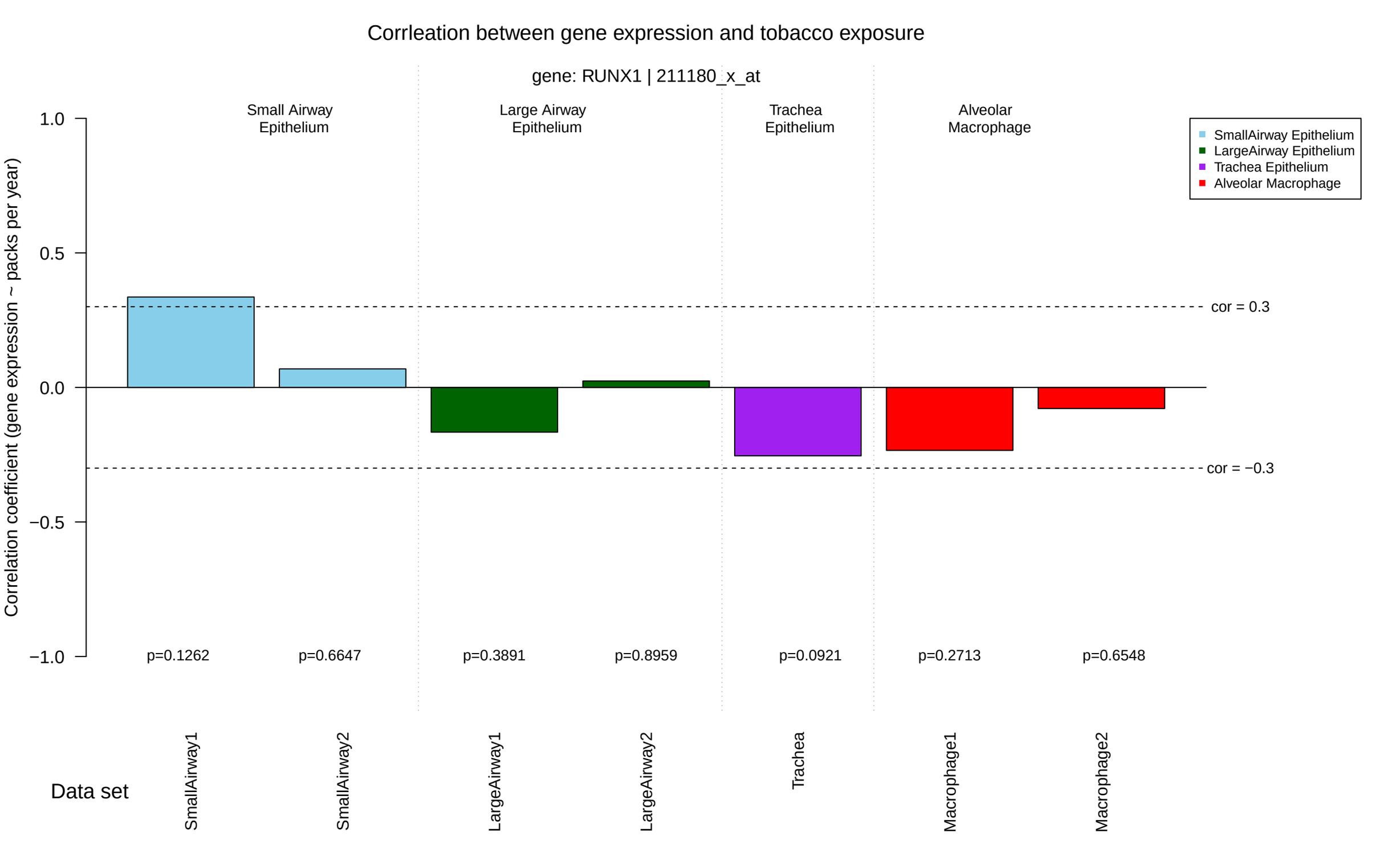

set: LargeAirway1 (p=0.5036)
RUNX1 | 211180_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1685)
RUNX1 | 211180_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0123)
RUNX1 | 211180_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8548)
RUNX1 | 211180_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9722)
RUNX1 | 211180_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RUNX1 | 211180_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1262
p=0.6647
p=0.3891
p=0.8959
p=0.0921
p=0.2713
p=0.6548
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2